

DevOps Lab Documentation

Program 4: Integrate Git with Docker for Source-Controlled Application Builds

Program-4

—>Dockerfile

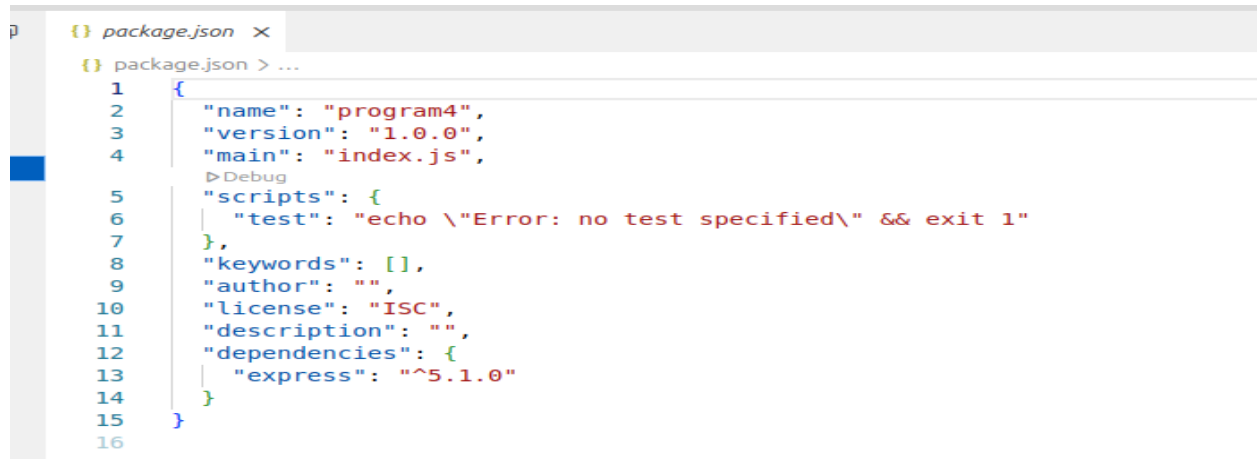
—>index.js

—>package.json

—>package-lock.json

S1. Create a file package.json

npm init -y



```
{  
  "name": "program4",  
  "version": "1.0.0",  
  "main": "index.js",  
  "scripts": {  
    "test": "echo \\\"Error: no test specified\\\" && exit 1"  
  },  
  "keywords": [],  
  "author": "",  
  "license": "ISC",  
  "description": "",  
  "dependencies": {  
    "express": "^5.1.0"  
  }  
}
```

S2. Create the file [index.js](#)



```
const express=require('express')  
const app=express()  
const PORT=3000  
app.get('/',(req,res)=>{  
  res.send("Docker + Github")  
});  
app.listen(PORT,()=>{  
  console.log(`Server runs on port:${PORT}`)  
});
```

S3. Add the dependency npm install express

```
package-lock.json x
package-lock.json > ...
1 {
2   "name": "program4",
3   "version": "1.0.0",
4   "lockfileVersion": 3,
5   "requires": true,
6   "packages": {
7     "": {
8       "name": "program4",
9       "version": "1.0.0",
10      "license": "ISC",
11      "dependencies": {
12        "express": "^5.1.0"
13      }
14    },
15    "node_modules/accepts": {
16      "version": "2.0.0",
17      "resolved": "https://registry.npmjs.org/accepts/-/accepts-2.0.0.tgz",
18      "integrity": "sha512-5cv6CtKwfgdmVqY1WIiXKc3Q1bkRqGLi+2W/6ao+6Y7gu/RCwRuAhGEzh5B4KlszSuTLgZYuqFqo5bImjNKng==",
19      "dependencies": {
20        "mime-types": "^3.0.0",
21        "negotiator": "^1.0.0"
22      },
23      "engines": {
24        "node": ">= 0.6"
25      }
26    }
27  }
28 }
```

S4. Create a Dockerfile

```
Dockerfile x
Dockerfile
1 #Stage1:
2 FROM node:18 AS build
3 WORKDIR /app
4 COPY package*.json
5 RUN npm install
6 COPY . .
7
8 #Stage2:
9 FROM node:18-slim
10 WORKDIR /app
11 COPY package*.json
12 COPY --from:build /app /app
13 EXPOSE 3000
14 CMD ["node", "index.js"]
```

S5. Initialize Git repository locally

git init

git add .

git commit -m "Initial commit"

git branch -M main

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
create mode 100644 node_modules/unpipe/README.md
create mode 100644 node_modules/unpipe/index.js
create mode 100644 node_modules/unpipe/package.json
create mode 100644 node_modules/vary/HISTORY.md
create mode 100644 node_modules/vary/LICENSE
create mode 100644 node_modules/vary/README.md
create mode 100644 node_modules/vary/index.js
create mode 100644 node_modules/vary/package.json
create mode 100644 node_modules/wrappy/LICENSE
create mode 100644 node_modules/wrappy/README.md
create mode 100644 node_modules/wrappy/package.json
create mode 100644 node_modules/wrappy/wrappy.js
create mode 100644 package-lock.json
create mode 100644 package.json
1rv24mc113_varun@varun-srivastava:~/DevOpslab/program4$ git branch -M main
```

S5. Link to remote repository

git remote add origin <http://github.com/varunvsSrivastava/program4.git>

S6. Push the code to Github

git push -u origin main

```
1rv24mc113_varun@varun-srivastava:~/DevOpslab/program4$ git remote add origin https://github.com/varunvsSrivastava/program4.git
1rv24mc113_varun@varun-srivastava:~/DevOpslab/program4$ git push -u origin main
Enumerating objects: 716, done.
Counting objects: 100% (716/716), done.
Delta compression using up to 8 threads
Compressing objects: 100% (661/661), done.
Writing objects: 100% (716/716), 722.58 KiB | 4.95 MiB/s, done.
Total 716 (delta 141), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (141/141), done.
To https://github.com/varunvsSrivastava/program4.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.
```

main 1 Branch 0 Tags

Go to file

Add file

Code

Ab

varunvsSrivastava program-4 fbd6d57 · 2 months ago 1 Commit

node_modules	program-4	2 months ago
Dockerfile	program-4	2 months ago
index.js	program-4	2 months ago
package-lock.json	program-4	2 months ago
package.json	program-4	2 months ago

lab

☆

🔍

🔗

Re

No

Cre